

SHUHAN TAN

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EDUCATION

University of Texas at Austin

Aug. 2021 -

Ph.D. in Computer Science

Research Assistant, GPA: 3.89/4.00

Sun Yat-Sen University

Aug. 2016 - Jun. 2021

B.Eng. in Computer Science & Technology

Major GPA: 93.1/100, Ranking: 2/189 (Top 1.1%)

PUBLICATIONS

- SceneGen: Learning to Generate Realistic Traffic Scenes.
Shuhan Tan*, Kelvin Wong*, Shenlong Wang, Sivabalan Manivasagam, Mengye Ren, Raquel Urtasun.
CVPR 2021.
- Class-imbalanced Domain Adaptation: An Empirical Odyssey.
Shuhan Tan, Xingchao Peng, Kate Saenko.
ECCV 2020 TASK-CV Workshop.
- LidarSIM: Realistic LiDAR Simulation by Leveraging the Real World.
Sivabalan Manivasagam, Shenlong Wang, Kelvin Wong, Wenyuan Zeng, Mikita Sazanovich, **Shuhan Tan**, Bin Yang, Wei-Chiu Ma, Raquel Urtasun.
CVPR 2020, Oral.
- Biomarker Localization by Combining CNN Classifier and Generative Adversarial Network.
Rong Zhang, **Shuhan Tan**, Ruixuan Wang, Siyamalan Manivannan, Wei-Shi Zheng.
MICCAI 2019.
- Weakly Supervised Open-set Domain Adaptation by Dual-domain Collaboration.
Shuhan Tan, Jiening Jiao, Wei-Shi Zheng.
CVPR 2019.

PATENT

- Systems and Methods for Simulating Traffic Scenes. **Shuhan Tan**, Kelvin Ka Wing Wong, Shenlong Wang, Sivabalan Manivasagam, Mengye Ren, Raquel Urtasun. United States Provisional Application No. 17/528,277.

AWARDS

SenseTime Scholarship (21 out of all AI-focus undergraduate students in China)

Dec. 2020

Excellent Student Scholarship, SYSU (top 5%, Twice)

Aug. 2016 - Jul. 2018

RESEARCH EXPERIENCE

Fairness of Deep Generative Models

Sep. 2020 - Mar. 2021

Advisor: Prof. **Bolei Zhou**

The Chinese University of Hong Kong

- Aims to offset the biases of a pre-trained GAN model without retraining.
- Proposed a post-processing method with GAN interpretation, which supports the sampling of latent codes that produce images with fairer attribute distributions. Experiments show that our approach can substantially improve the fairness of image generation.

Simulation for Autonomous Driving

Oct. 2019 - Aug. 2020

Advisors: Prof. **Raquel Urtasun**, Prof. **Shenlong Wang**

Uber ATG

Topic 1: Realistic Traffic Scene Generation.

- Aims to generate realistic traffic scenes that can be used to train or test of self-driving tasks.
- Proposed a data-driven approach to sequentially generate traffic actors to form realistic traffic scenes given map layout. Our method is able to automatically generate realistic traffic scenes across at large scale, and used to train 3D perception model that generalize to the real world.
- Publication: **CVPR**, 2021.

Topic 2: Realistic LiDAR Simulation.

- Aims to build realistic LiDAR simulator and its testing methods for self-driving simulation.
- Designed a method to simulate realistic and rare driving scenes using LiDAR Simulator and 3D assets. Created scenes are useful for efficient edge-case testing for autonomous driving.
- Publication: **CVPR**, 2020.

Class-Imbalanced Domain Adaptation

July. 2019 - Sep. 2019

Advisor: Prof. **Kate Saenko**

Boston University

- Aims to remove impractical assumptions to achieve a more practical domain adaptation method.
- Proposed the Class-imbalanced domain adaptation problem: how to conduct domain adaptation under both feature and label distribution gaps. Built a large benchmark for this problem with 22 cross-domain tasks. Experiments on state-of-the-art methods reveal that most existing methods fail to handle this problem. Proposed a new method that consistently outperforms current methods.
- Publication: **ECCV TASK-CV Workshop**, 2020.

Domain Adaptation and Medical Imaging

Jan. 2018 - June. 2019

Advisor: Prof. **Wei-Shi Zheng**

Sun Yat-Sen University

Topic 1: Weakly Supervised Domain Adaptation.

- Aims to utilize domain adaptation for problems where we have partially-labeled data domains.
- Proposed the weakly-supervised open-set domain adaptation setting, where we relax the conventional assumption that there exists a fully-labeled source domain. Designed a novel algorithm to transfer knowledge bilaterally between two partly-labeled domains, which was state-of-the-art on both standard datasets and a real-world Person Re-identification.
- Publication: **CVPR**, 2019.

Topic 2: Weakly Supervised Biomarker Localization.

- Aims to predict pixel-level biomarkers locations in medical images with only image-level labels.
- Proposed to combine a CNN classifier and a GAN such that CNN classifier and the GAN's discriminator can effectively help the encoder-decoder in the GAN to remove biomarkers.
- Publication: **MICCAI**, 2020.